Create a simple Dockerfile for a Python Flask application.
Build the Docker image locally and tag it appropriately for GitHub Container Registry (ghcr.io).

```
(base) ┌──(ravan666㉿kali)-[~]
└─$ cd Desktop

(base) ┌──(ravan666㉿kali)-[~/Desktop]
└─$ mkdir flask-docker

(base) ┌──(ravan666㉿kali)-[~/Desktop]
└─$ cd flsak-docker
cd: no such file or directory: flsak-docker

(base) ┌──(ravan666㉿kali)-[~/Desktop]
└─$ cd flask-docker

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ nano app.py

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ nano requirements.txt

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ nano Dockerfile

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ docker build -t flask-ghcr .
[+] Building 9.3s (11/11) FINISHED                                      docker:default
 => [internal] load build definition from Dockerfile                                0.0s
 => => transferring dockerfile: 200B                                                0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                 8.5s
 => [auth] library/python:pull token for registry-1.docker.io                       0.0s
 => [internal] load .dockerignore                                                   0.0s
 => => transferring context: 2B                                                     0.0s
 => [1/5] FROM docker.io/library/python:3.11-slim@sha256:9358444059ed78e2975ada2c189f1c1a3144a5dab6f3  0.1s
 => => resolve docker.io/library/python:3.11-slim@sha256:9358444059ed78e2975ada2c189f1c1a3144a5dab6f3  0.1s
 => [internal] load build context                                                   0.1s
 => => transferring context: 460B                                                   0.0s
 => CACHED [2/5] WORKDIR /app                                                       0.0s
 => CACHED [3/5] COPY requirements.txt .                                            0.0s
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt                 0.0s
 => [5/5] COPY . .                                                                  0.1s
 => exporting to image                                                              0.4s
 => => exporting layers                                                             0.1s
 => => exporting manifest sha256:f90e1bd3b0765e90c95ffedad936a9064eadbc89f2883091aa55cfcd2623a980  0.0s
 => => exporting config sha256:1a0c8e5044aaa811c98aedc01636f18886c6a8e0bb4a924a8ad90f5d0d5b7ca2  0.0s
 => => exporting attestation manifest sha256:78017af93633753e890d96c30f68e8d501cc7d6c384944df7c16ae03  0.1s
 => => exporting manifest list sha256:231d72f6621f87e6a6c9e7265d37f593b6c0a557329d1492c03d231c168fe9e  0.0s
 => => naming to docker.io/library/flask-ghcr:latest                                0.0s
 => => unpacking to docker.io/library/flask-ghcr:latest                             0.0s

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ docker images
                                                  i Info →   U  In Use
IMAGE                                   ID             DISK USAGE   CONTENT SIZE   EXTRA
calculator:1.0                          744c79e8b26b       675MB         211MB    U
docker-demo:1.0-SNAPSHOT                9117215bcf7c       675MB         211MB    U
flask-app:latest                        c5abcca3db46       210MB        51.1MB    U
flask-ghcr:latest                       231d72f6621f       210MB        51.1MB
ghcr.io/sonalmehta1258/ghcr-demo:latest 97f5b6b643f0      92.5MB          26MB
hello-world:latest                      05813aedc15f      25.9kB        9.52kB    U
mongo:6                                 03cda579c8ca      1.06GB         273MB
mysql:5.7                               4bc6bc963e6d       700MB         149MB
```

```
Login did not succeed, error: Error response from daemon: Get "https://ghcr.io/v2/": denied: denied
Username (sonalmehta1258): ^C

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ docker logout
Removing login credentials for https://index.docker.io/v1/

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ docker logout ghcr.io
Removing login credentials for ghcr.io

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ echo YOUR_PAT | docker login ghcr.io -u sonalkrmehta --password-stdin
Error response from daemon: Get "https://ghcr.io/v2/": denied: denied

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ docker login

USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login -u <username>'


Your one-time device confirmation code is: WKBL-GGBN
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser…

Login Succeeded

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ echo YOUR_PAT | docker login ghcr.io -u sonalkrmehta --password-stdin
Error response from daemon: Get "https://ghcr.io/v2/": denied: denied

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ echo ghp_fdNHf8kxcxBkA74YHTrVeUhxzA75ky4KFepk | docker login ghcr.io -u sonalkrmehta --password-stdin
Error response from daemon: Get "https://ghcr.io/v2/": context deadline exceeded

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ echo ghp_fdNHf8kxcxBkA74YHTrVeUhxzA75ky4KFepk | docker login ghcr.io -u sonalkrmehta --password-stdin

Error response from daemon: Get "https://ghcr.io/v2/": net/http: request canceled while waiting for connection (Client.Timeout exceeded while awaiting headers)

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ echo ghp_fdNHf8kxcxBkA74YHTrVeUhxzA75ky4KFepk | docker login ghcr.io -u sonalkrmehta --password-stdin

Login Succeeded

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ docker images
                                                  i Info →   U  In Use
IMAGE                                   ID             DISK USAGE   CONTENT SIZE   EXTRA
calculator:1.0                          744c79e8b26b       675MB         211MB    U
docker-demo:1.0-SNAPSHOT                9117215bcf7c       675MB         211MB    U
flask-app:latest                        c5abcca3db46       210MB        51.1MB    U
flask-ghcr:latest                       231d72f6621f       210MB        51.1MB    U
ghcr.io/sonalmehta1258/ghcr-demo:latest 97f5b6b643f0      92.5MB          26MB
hello-world:latest                      05813aedc15f      25.9kB        9.52kB    U
mongo:6                                 03cda579c8ca      1.06GB         273MB
```

```
ravan666@kali: ~/Desktop/flask-docker
Session  Actions  Edit  View  Help
Username (sonalmehta1258): Password:
^C



time="2026-04-03T10:20:00+05:30" level=info msg="Error logging in to endpoint, trying next endpoint" endpoint="{https://ghcr.io 0xc00038a000}" error="Get \"https://ghcr.io/v2/\": denied: denied"
Get "https://ghcr.io/v2/": denied: denied

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$

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└─$

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ ghcr.io/sonalkrmehta/flask-ghcr:latest
zsh: no such file or directory: ghcr.io/sonalkrmehta/flask-ghcr:latest

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ docker run -p 5000:5000 flask-ghcr
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [03/Apr/2026 04:54:14] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [03/Apr/2026 04:54:14] "GET /favicon.ico HTTP/1.1" 404 -
^C

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ ghcr.io/sonalmehta1258/flask-ghcr:v1
zsh: no such file or directory: ghcr.io/sonalmehta1258/flask-ghcr:v1

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ ghcr.io/sonalmehta1258/flask-ghcr:latest
zsh: no such file or directory: ghcr.io/sonalmehta1258/flask-ghcr:latest

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ docker login ghcr.io
Authenticating with existing credentials... [Username: sonalmehta1258]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login did not succeed, error: Error response from daemon: Get "https://ghcr.io/v2/": denied: denied
Username (sonalmehta1258): ^C

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ docker logout
```

```
ravan666@kali: ~/Desktop/flask-docker
Session  Actions  Edit  View  Help

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ echo 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 | docker login ghcr.io -u sonalkrmehta --password-stdin
Error response from daemon: Get "https://ghcr.io/v2/": context deadline exceeded

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ echo 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 | docker login ghcr.io -u sonalkrmehta --password-stdin

Error response from daemon: Get "https://ghcr.io/v2/": net/http: request canceled while waiting for connection (Client.Timeout exceeded while awaiting headers)

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ echo 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 | docker login ghcr.io -u sonalkrmehta --password-stdin

Login Succeeded

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ docker images
                                                                                                                    i Info →   U  In Use
IMAGE                                   ID             DISK USAGE   CONTENT SIZE   EXTRA
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docker-demo:1.0-SNAPSHOT                9117215bcf7c       675MB          211MB     U
flask-app:latest                        c5abcca3db46       210MB         51.1MB     U
flask-ghcr:latest                       231d72f6621f       210MB         51.1MB     U
ghcr.io/sonalmehta1258/ghcr-demo:latest 97f5b6b643f0      92.5MB           26MB
hello-world:latest                      05813aedc15f      25.9kB         9.52kB     U
mongo:6                                 03cda579c8ca      1.06GB          273MB
mysql:5.7                               4bc6bc963e6d       700MB          149MB
nginx:latest                            bc45d248c4e1       240MB         65.8MB
node_mongo-nodeapp:latest               9b8b122aec21      1.63GB          406MB
python-app:latest                       5caad79b6c86       203MB         49.7MB     U
redis:latest                            1c854d54ecd1       206MB         55.3MB
skm:latest                              6f5e8b4299b8       200MB         48.8MB     U
sys-monitor:latest                      64adc1e250c1       204MB         49.7MB     U
wordpress:latest                        ffef0dca1f0f      1.08GB          275MB

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ ghcr.io/sonalkrmehta/flask-ghcr:latest
zsh: no such file or directory: ghcr.io/sonalkrmehta/flask-ghcr:latest

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ docker tag flask-ghcr ghcr.io/sonalkrmehta/flask-app:v1

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ echo 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 | docker login ghcr.io -u sonalkrmehta --password-stdin

Login Succeeded

(base) ┌──(ravan666㉿kali)-[~/Desktop/flask-docker]
└─$ docker push ghcr.io/sonalkrmehta/flask-app:v1
The push refers to repository [ghcr.io/sonalkrmehta/flask-app]
d5ee0ff07539: Pushed
2b761b452cee: Pushed
7524aba74b8c: Pushed
6944007a5b45: Pushed
df80f2df6bee: Pushed
ec781dee3f47: Pushed
ea4b2e75d902: Pushed
ba1de931814a: Pushed
e35b92e2619d: Pushed
v1: digest: sha256:231d72f6621f87e6a6c9e7265d37f593b6c0a557329d1492c03d231c168fe9e1 size: 856
```

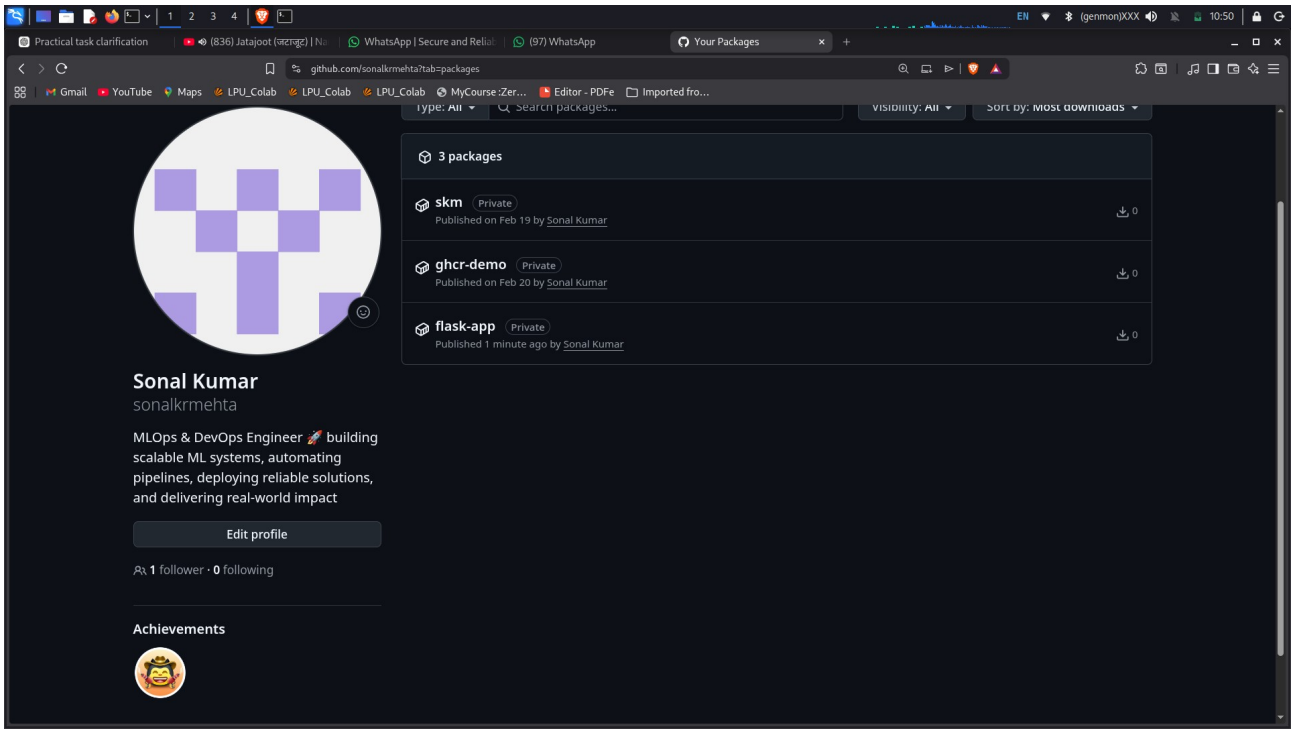
Practical task clarification
(836) Jatajoot (जटाजूट) | Na
WhatsApp | Secure and Reliab
(97) WhatsApp
Your Packages
github.com/sonalkrmehta?tab=packages
Gmail
YouTube
Maps
LPU_Colab
LPU_Colab
LPU_Colab
MyCourse :Zer...
Editor - PDFe
Imported fro...
Type: All
Search packages...
Visibility: All
Sort by: Most downloads
3 packages
skm
Private
Published on Feb 19 by Sonal Kumar
ghcr-demo
Private
Published on Feb 20 by Sonal Kumar
flask-app
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